M13-C3 — H3 BK-leading distractor
ECAI target c retains isolated A

H3 run / analysed / not a Bucket 3 claim

1 Status

H3 run / analysed. ECAI (primary) and AAMAS (secondary) on fixture `m13_bucket3_binary_bd_and_lead_a` / `n30_seed42` are complete. Markdown twin: `h3_bk_leading_distractor.md`.

Lead finding: BK-leading isolated a is pulled into ECAI's theory for deterministic child c (first fold selects `a_val_*`; delta conditioned on a; not the evaluator BD AND). **Secondary:** brave `solved` can rely on per-row assumption choice where BK underdetermines the target. H0–H2 remain closed. **Not a Bucket 3 claim.** H4 cautious infrastructure is ready but unrun; H5 cautious Greedy is deferred.

2 Research question

On a BD AND collider with BK-leading independent isolated root A, does ECAI learning of target c select a in the first fold and retain a in the final ABA framework rather than returning the evaluator-only rule `c(A) :- b_val_1(A), d_val_1(A).`?

3 Main finding (lead)

On frozen sample `n30_seed42`, ECAI **solves** target c, but BK-leading isolated a is pulled into the learned theory.

Decisive first fold:

```
folding result: c(A) <- [a_val_1(A)]
```

then assumption introduction of α_1. Both ordinary target rules remain conditioned on a. Emitted delta:

```
c(A) :- alpha_1(A), a_val_1(A).
c(A) :- alpha_2(A), a_val_0(A).
c_alpha_1(A) :- b_val_0(A).
c_alpha_1(A) :- alpha_3(A), b_val_1(A).
c_alpha_2(A) :- d_val_0(A).
c_alpha_2(A) :- b_val_0(A).
c_alpha_3(A) :- alpha_1(A), a_val_1(A).
```

(plus `assumption`/`contrary` declarations).

Object	Content
Evaluator-only reference for c	`c(A) :- b_val_1(A), d_val_1(A).` (**no** a)
ECAI metrics body variables for c	$\{a\}$
ECAI delta	assumption-rich; both target rules involve a
Among C=1 on n=30	all 15 have $(b, d) = (1, 1)$; both values of a (10/5)

Distractor inclusion / **first-fold distraction** away from the mechanism-aligned B, D AND. `solved` + audit SAT does **not** mean the evaluator BD rule was recovered.

4 Secondary finding (brave residual)

ECAI brave `solved` can be sample-adequate via **per-row assumption choice** even when BK features do not determine the target. On H3 ECAI c, this is concentrated in the nested a=1 / α_1–α_3 repair. The a=0 / `c_alpha_2` side is more BK-readable. Clearer prior exemplar: H0 ECAI target a, rows 9 vs 26 (`m13_bucket3_binary_collider_and`).

Bound: `solved` + **audit SAT $\neq$ BK$\rightarrow$target predictor recovered** / **mechanism-aligned.**

5 Fixture / setup

Fixture `m13_bucket3_binary_bd_and_lead_a`: $B \rightarrow C \leftarrow D$; isolated $A \sim$ Bernoulli(1/2); $B \sim$ Bernoulli(4/5); $D \sim$ Bernoulli(7/10); $C = B \wedge D$. For target c, BK predictors are a, then b, then d (leading distractor). Evaluator reference: `c(A) :- b_val_1(A), d_val_1(A).`. Primary arm ECAI; secondary AAMAS; same frozen `n30_seed42`.

6 AAMAS contrast

Same sample. AAMAS solves c with

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_0(A), b_val_1(A), d_val_1(A).
```

so AAMAS also retains `a_val_*` (distractor omission failure; different shape from ECAI). Versus H2: trailing isolated d under AAMAS retained `d_val_*`; here leading isolated a is retained under both arms.

7 Secondary targets

ECAI: all of a, b, c, d `solved`. AAMAS: c `solved`; a, b, d `completed_no_solution`. Other targets are infrastructure retention only.

8 Boundaries / non-claims

- Not a Bucket 3 claim.

- Inclusion of a is not causal recovery of an $A \to C$ edge.
- `solved` / SAT is not automatically mechanism-aligned.
- "ECAI always solves" is not the H3 question.
- H0–H2 unchanged except for cross-links. H4/H5 have no result.

9 Next

H3 documentation is complete. H4 cautious target-wise infrastructure is ready, but no H4 collection has been run; H5 remains deferred. Still **no Bucket 3 claim**.